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BACTERIA FIGHTING PAPER

Scientists at the Bar-Ilan University, Israel have developed paper which can fight bacteria. Each paper is coated with silver nanoparticles which penetrates the paper surface to a depth of more than 1 micrometer. The paper can fight bacteria such as *E. coli* and *S. aureus*. This invention will surely change the meaning of paper soaps.

Read more

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THE YEAR IN TECH

Year ending lists are always fascinating. Daniel Raffel, formerly a product manager with Yahoo! compiles the most comprehensive and simply awesome list of favourite new products, projects and features of 2010. If its not on this list it probably wasn't important. A definite must read for any tech enthusiast, complete with insightful opinions from the author.

<http://tcn.ch/dKqe97>

elsewhere on the floor of the case. A few cases do have external 2.5-inch bays, but these are very rare—most of the time, the case will provide some method of securing a 2.5-inch drive in a 3.5-inch bay.

Check Your Expansion Slots

Most ATX cases will have six or seven slots on the rear panel for adding expansion cards; microATX cases will usually have four. We have, however, seen a few cases that buck these trends—high-end gaming cases designed for users of multiple two-slot video cards can have as many as 10 slots! For the most part, if the form factor of your motherboard matches that of your case, you'll be fine. But check the specs when you're buying, just in case.

Tool-Free?

Once upon a time, it was taken for granted that doing anything inside a PC required a Phillips screwdriver. Not anymore. While it's still a must-have tool, many cases these days have tool-free assemblies that let you access the inside of the case and secure most components without ever needing the screwdriver at all. (The exception is the motherboard; you'll always need to secure that with screws.) Different cases have different methods, but common ones include plastic clips for the optical and hard drives, specialized rivets that let the hard drives slide in and out of their bays, a press-until-click rail for the expansion cards, and so on. Better (read "more expensive") cases usually have better tool-free systems, but we find that even the best of them are more frustrating than the old-fashioned Phillips way.

PSU: Built-In or not?

Should you buy a case with a built-in power supply? It depends. The cheaper the case



and the power supply together, the worse an idea it probably is—low-end power supplies can be inefficient and shorter-lived, which means you wouldn't be saving money in the long run. If, on the other hand, you see a special case-PSU-combo deal from a major manufacturer like Antec, Cooler Master, or Thermaltake, you should be fine. Those companies are in the business of both cases and power supplies, so you can feel better about the pairings they provide. You may pay a little more, but the extra peace of mind will be worth it.

Keep It Cool

The biggest way that a case can contribute to—or detract from—a system's performance is in how well it cools (or doesn't cool) the interior components. This means you'll want to pay attention to what sort of fans your case has out of the box—and how many you can add to it later. Just about all computers have an exhaust fan on the rear panel for expelling heated air, and most have intake fans on the front panel for bringing air into the case. (Frequently, the intake fan is placed right near the hard drives, so they can be cooled before anything else.) The more

powerful the system, the more additional fans you're likely to see; high-performance PCs can also include extra exhaust fans in the case's ceiling, or intake fans in the side panel (not coincidentally, those are often located right over the processor). The thing to remember, though, is that the more fans you have, and the smaller they are, the more noise your computer will produce.

Does Form and Function Matter?

We're saving this for last because it's technically the least important—at least as far as what your system can or can't do—but the way a case looks almost always plays a role in whether you decide to buy it. Cases may be the easiest components to buy, because there's often no question about what you need, but they're also among the hardest because they're how you present your computer to the world. Do you like the case's shape and its colour? Are you okay with something that looks traditional, or are you hankering for something that looks like it was designed for a high-budget sci-fi movie? Do you want tons of individual LEDs and lighting strips adorning every surface, or would you prefer to see as little illumination as possible? All these and more can be had—for a price.

Cases can cost anywhere from under \$50 to over \$800 (and no, that's not a typo), but you can get cases with everything you need for well under \$100 and fully decked-out gaming cases for around \$200. The hardest part will be finding the one that best matches your personality and the system you're hoping to build—but that's also one of the most fun and rewarding parts of assembling a computer from scratch.

For details and other tips visit:

<http://bit.ly/g2Njrr>